

Hybrid privacy-aware semantic search: SVD-truncated document geometry and CKKS-encrypted query reranking under a restricted threat model

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Abstract

Dense vector embeddings underpin modern semantic search and Retrieval-Augmented Generation (RAG), yet a growing body of work shows that an embedding can be inverted back into the text that produced it with alarming fidelity: once a vector database leaks, the documents behind it leak with it. The two textbook defences lie at opposite extremes—encrypting the entire search with fully homomorphic encryption is cryptographically sound but far too slow for corpora of millions of documents, whereas injecting privacy-preserving noise degrades ranking quality long before it provides meaningful protection.

We study a middle path built on an asymmetry between the static document collection and the dynamic user query. The collection is protected geometrically: every document vector is compressed onto a lower-dimensional subspace and then rotated by a secret orthogonal transformation held only by the data owner. The query is protected cryptographically: it is processed under homomorphic encryption so that the server—which honestly follows the protocol but may inspect everything it stores—never observes the query values or the similarity scores it computes. The encryption parameters are fixed by a small, reproducible offline benchmark over the discrete CKKS grid rather than by manual trial and error.

Beyond the construction itself, we contribute mathematically proven theoretical results: we derive a tight lower bound on the reconstruction error of any attacker confined to the protected subspace, making precise how much information the geometric projection provably removes. Large-scale experiments on a one-million-document corpus and five contemporary text encoders corroborate the empirical side—the protection preserves retrieval quality and, on the strongest encoders, even slightly improves it by acting as a linear denoiser; query latency stays comfortably below one second; and an off-the-shelf text-reconstruction attack confined to the protected space collapses to the noise floor. We then deliberately step outside that comfortable regime and measure what happens against stronger adversaries: a known-plaintext attacker recovers the secret rotation by orthogonal Procrustes from roughly as many leaked pairs as the retained dimension, the public product-quantization codes preserve most nearest-neighbour structure, and a calibrated-noise and random-projection baseline suite positions the trade-off against natural alternatives; a BEIR check shows the truncation is an encoder-dependent accuracy cost rather than a free denoiser on graded-relevance tasks. We therefore state the boundaries plainly: query confidentiality is cryptographic, but document protection rests on SVD truncation and a secret rotation that are an empirical obfuscation layer—not a cryptographic primitive—and we delimit precisely the threat model under which each claim holds.

Keywords: homomorphic encryption, CKKS, privacy-preserving information retrieval, semantic search, embedding inversion, non-adaptive attacker, restricted threat model, SVD truncation, product quantization, retrieval-augmented generation.

1 Introduction

The privacy of vector representations of text—“embeddings”—has become a first-class concern in production retrieval systems. State-of-the-art encoders (Sentence-BERT [1], multilingual-E5 [2], GTR [3], BGE-M3 [4]) push the per-document representation through a $\sim 10^2$ -dimensional bottleneck that nevertheless preserves enough information to reconstruct the original text with high BLEU. VEC2TEXT [7] reports BLEU $\approx 97.3\%$ on 32-token inputs when the attacker has query access to the encoder; GEIA [8] and TEIA [9] show that even snapshot attacks (an adversary with read access to the vector database) recover personally identifiable information at high rates. For Retrieval-Augmented Generation pipelines whose vector stores typically contain customer support tickets, internal corporate documents and personal data, this rule of thumb—“a leak of the embedding is a leak of the text”—is now well documented [14].

The defensive landscape is bracketed by two extremes that, individually, do not satisfy industrial requirements. Differential privacy [18, 19] adds calibrated noise per-record but, in the naive coordinate-Gaussian formulation on retrieval-quality embeddings, destroys ranking quality long before the privacy budget reaches a level considered useful. Fully homomorphic encryption [23, 28] allows arbitrary computation on ciphertexts, but the ciphertext-ciphertext (ct-ct) regime is too expensive for top- k search across 10^6 documents.

Nature and positioning of this work. This is a *scientific study*, not a systems-engineering paper. SVD truncation, random orthogonal rotation, product quantisation and CKKS are individually well known; we do not claim a new system or a deployable product. The research questions are: **(RQ1)** what does projection provably remove—is there a formal lower bound on what any projection-restricted decoder can reconstruct? **(RQ2)** how do SVD truncation and an unknown secret rotation jointly reshape the privacy–accuracy–latency trade-off of semantic search at 10^6 -document scale? **(RQ3)** does data-driven SVD truncation behave as a destructive lossy step, or—as we find—as a linear *denoiser* for retrieval-trained encoders? Accordingly, the secret rotation is studied as an empirical obfuscation phenomenon, not proposed as a cryptographic primitive, and the formal lemma is a projection bound, not an inversion-security theorem.

Object of study and scientific contribution. The object we analyse is an *asymmetric* hybrid construction (described below); the contribution is its theoretical and empirical characterisation under a credible—but explicitly restricted—threat model, not its engineering realisation:

- For the *static* side (the database), we apply a deterministic SVD truncation V_k with $k = d/2$, immediately followed by a Haar-uniform secret orthogonal rotation $R \in O(k)$. The truncation is chosen so that the proxy criterion $\sigma_{rec} \geq 0.10$ (Section 6) is met uniformly for all tested encoders; the rotation hides the orientation of the basis from any attacker that does not know R . We then publish a Product-Quantisation artefact (codebook + per-document PQ codes), trained *in the rotated space*, so that stage-1 candidate filtering can be performed locally by the client without sending the query to the server.
- For the *dynamic* side (the query), we apply CKKS encryption. The protocol is engineered so that the server-side reranking over the short-list of K_{cands} candidates only needs the ciphertext-plaintext (ct-pt) mode—the costly relinearisation step specific to ct-ct multiplication is avoided—bringing the cryptographic budget for one query well below one second.
- The CKKS parameters $(N_{poly}, \log_2 Q, d_{proj}, \Delta)$ are fixed by a small, reproducible offline micro-benchmark over the discrete parameter grid (Section 5): we keep the minimum-latency configuration that satisfies the conservative security tables and an accuracy tolerance. The selected configuration $(N_{poly} = 8192, [60, 40, 60], \Delta = 2^{40})$ yields a $\approx 1.7\times$ speed-up over the TenSEAL stock setting $[60, 40, 40, 60]$ at the same parameter-table security bound. This is a deterministic engineering choice over a published grid; we make no machine-learning claim.

Notation. To avoid the overloading of the symbol N , we use N_{poly} for the CKKS polynomial-modulus degree, N_{docs} for the corpus size, and K_{cands} for the stage-1 short-list length throughout.

Theorem and an empirical link. We prove a tight lower bound on the L_2 reconstruction error of any decoder whose image is constrained to $\text{span}(V_k)$ (Lemma 1). The bound makes σ_{rec} a meaningful proxy of the difficulty an inverter faces; its empirical relation to the BLEU of an off-the-shelf VEC2TEXT attack is then formalised as Empirical Hypothesis 1 and tested on GTR-BASE embeddings.

Roadmap. Section 2 reviews the related literature; Section 3 introduces the threat model, separating *document*, *query* and *access-pattern* privacy, and gives both a coverage table and a claims-vs-evidence table; Section 4 describes the asymmetric architecture, including a cryptographic-reproducibility subsection; Section 5 states the reproducible CKKS parameter selection; Section 6 proves the projection-decoder lemma and delimits its scope; Section 7 presents four experiments; Section 9 discusses limitations, required attack experiments and open work.

2 Related Work

Embedding inversion. VEC2TEXT [7] trains an encoder-decoder model that iteratively rewrites a candidate hypothesis until its embedding matches a target vector; on 32-token MS-MARCO inputs the attack achieves $\text{BLEU} \approx 97.3\%$ and $\approx 92\%$ exact recovery. GEIA [8] instead trains a generative decoder that maps a target embedding to a prompt and reconstructs the whole sentence in a single pass. TEIA [9] removes the need to query the victim encoder, training a surrogate inverter that transfers across embedding models. Adjacent work on membership-inference [16] and on extracting training texts from large language models [17] confirms that the risk surface of dense retrievers is broader than the embedding-inversion literature alone suggests.

Modern inversion attacks (2024–2026). Since VEC2TEXT the threat has strengthened along two axes that matter for any geometric obfuscation. *Few-shot alignment* attacks such as ALGEN [10] learn a linear map between embedding spaces from $\sim 10^3$ leaked pairs and reach VEC2TEXT-level recovery without query access to the victim encoder; *zero-shot* attacks such as ZSINVERT [11] invert with no encoder-specific training; the unsupervised translator VEC2VEC [12] aligns two embedding spaces with no paired data at all and argues that a leaked vector store should be treated as plaintext; and ZERO2TEXT [13] updates an online token-by-token regression. These attacks are *alignment-based* or *adaptive* and exploit the universal geometry of embedding spaces, so a single secret rotation—which only hides the orientation of a fixed basis—does not by itself neutralise them. They therefore lie outside the non-adaptive threat model under which our construction is analysed (Section 3); we quantify the closest members of this family—known-plaintext linear alignment and an aligned off-the-shelf inverter—directly in Section 8, and flag the fully learned variants as the main residual gap in Section 10.

Differential privacy for embeddings. Lyu et al. [20] investigate per-coordinate Gaussian noise on dense representations; the empirical finding is that retrieval quality degrades sharply long before ε reaches a useful range. Random-projection-based privacy [22, 21] predates modern dense retrieval; the data-dependent SVD projection used here is closer to a noise-removal technique than to a privacy-preserving DP mechanism, although σ_{rec} provides a useful proxy quantity.

Homomorphic encryption. The CKKS scheme [23] introduced approximate-arithmetic FHE; OpenFHE [28] and TenSEAL provide production-grade implementations. The bootstrapping cost [24] pushes practitioners to leveled circuits; specialised compilers [29, 30] optimise scale and

modulus chains. GPU acceleration of CKKS bootstrap [31] and Intel HEXL [32] have made larger leveled circuits feasible.

Hybrid privacy-preserving retrieval. Tiptoe [34] combines clustering with PIR for query privacy. SealPIR [35], OnionPIR [37] and SimplePIR [36] provide single-server PIR with practical query sizes. The architecture in this paper is complementary: ct-pt CKKS reranking takes care of the value side of the leakage, while access patterns remain a separate problem for which PIR/ORAM-like primitives are an appropriate composition.

3 Threat Model

We adopt the asymmetric threat model *trusted client / honest-but-curious server / non-adaptive attacker / unknown rotation*, characterised as follows.

- The *client* is the data owner. It runs the encoder E , generates the secret keys $\mu, V_k, R, sk_{\text{CKKS}}$ and never shares them. The client is trusted in software and key management.
- The *server* runs the search service over the rotated database $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ and the public PQ artefact. It is honest-but-curious: it follows the protocol but attempts to extract textual information from the data it sees and from the queries it processes.
- The *attacker* is non-adaptive: it uses an off-the-shelf VEC2TEXT model trained without prior knowledge of R . This is the model under which the construction is *analysed*. We do not stop there: Section 8 steps outside it and empirically measures known-plaintext recovery of R , public-PQ-code leakage and an aligned off-the-shelf inverter, so the boundary of the non-adaptive model is quantified rather than merely asserted. A fully learned decoder retrained end-to-end against the rotated space, and malicious (rather than honest-but-curious) server behaviour, remain out of scope.

Three distinct privacy notions. A recurring source of confusion in hybrid designs is the conflation of *what* is protected. We therefore separate three independent notions and state, for each, exactly which component is responsible:

- **Document privacy.** The plaintext rotated database E_{rot} is visible to the server. It is protected only by SVD truncation (σ_{rec} , a proxy quantity) and the secret rotation R (an empirical obfuscation layer). *This is not a cryptographic protection*; it fails under known- R , known-plaintext, or an adaptive decoder.
- **Query privacy.** CKKS hides the numerical value of the query vector and of the similarity scores from the server. This part *is* cryptographic, at the chosen security level.
- **Access-pattern privacy.** The candidate IDs reranked per query, and the public PQ codes, are *not* hidden. CKKS does not address this channel; it requires composition with a PIR/ORAM-style primitive.

In particular, the public PQ artefact (codebook + per-document codes, trained in the rotated space) is itself a lossy compressed view of E_{rot} . Whether neighbourhood, cluster or topic structure leaks from the public PQ codes alone is measured directly in Section 8 (Table 10): it does, and we treat the PQ artefact as an exposed channel accordingly.

Table 1 summarises the coverage and Table 2 maps each claim to the evidence that supports it. The coverage table is intentionally asymmetric: ct-pt reranking takes care of the value channel, while the access-pattern channel (which document IDs are reranked) is left open and would require composition with a PIR-style primitive, and the public-PQ-artefact and known-plaintext channels—now quantified in Section 8—remain exposed rather than defended.

Table 1: Threat coverage of the proposed protocol.

Attack vector	Covered?	Comment
Snapshot of E_{rot} without R (off-the-shelf Vec2Text)	Partial	Heuristic; empirically validated against non-adaptive Vec2Text.
Network interception, leakage of similarity scores	Yes	CKKS at tc128; sk never leaves the client.
Known R (known-rotation attacker)	No	Reduces to σ_{rec} from SVD truncation only.
Known-plaintext pairs ($\text{text}_i, E_{\text{rot},i}$)	No	R recovered by Procrustes from $\approx k$ pairs (Sec. 8.1, Table 8).
Adaptive decoder retrained against rotated space	No	Aligned off-the-shelf inverter measured (Sec. 8.4); a <i>learned</i> decoder is the main open case.
Malicious server (protocol deviation, substitution)	No	Verifiable computation / TEEs are an orthogonal addition.
Access-pattern leakage on candidate IDs	No	CKKS hides values, not access patterns; requires PIR/ORAM.
Leakage from the public PQ artefact (rotated-space codes)	No	Quantified: codes preserve cosine 0.95 and 67% of top-10 neighbours (Sec. 8.3, Table 10).

4 Method

The asymmetric defence is illustrated in Figure 1. The architecture is the union of an offline data-preparation phase (Section 4.1) and the online query protocol (Section 4.2).

4.1 Offline phase

Encoder and centring. The data owner runs the encoder E on a corpus $D = \{T_1, \dots, T_{N_{\text{docs}}}\}$ to obtain $X \in \mathbb{R}^{N_{\text{docs}} \times d}$ with L_2 -normalised rows. The global centroid $\mu \in \mathbb{R}^d$ is computed and the corpus is centred: $X_c = X - \mu$.

SVD truncation. A randomised SVD [38] $X_c \approx U_k \Sigma_k V_k^\top$ produces the orthonormal matrix $V_k \in \mathbb{R}^{d \times k}$ that spans the dominant k -dimensional subspace. We use $k = d/2$ in the canonical operating point: this is the smallest k at which the relative reconstruction error

$$\sigma_{\text{rec}}^2(E; V_k) = \frac{\|E - \pi_k(E)\|_F^2}{\|E\|_F^2} = 1 - \eta_k$$

exceeds the proxy threshold $\sigma_{\text{rec}} \geq 0.10$ uniformly across all five tested encoders (Section 7).

Secret rotation. A Haar-uniform random orthogonal matrix $R \in O(k)$ is generated by QR-decomposing a Gaussian matrix and correcting the diagonal of R to obtain a uniform distribution on the group. The protected document representation is

$$v'_i = T(v_i) = R V_k^\top (v_i - \mu) \in \mathbb{R}^k.$$

Public PQ artefact. A faiss IndexPQ index is trained *in the rotated space* E_{rot} with $M = k/4$ subquantisers and 8 bits per code. Both the codebook and the per-document codes are public; the client downloads them once at on-boarding. Training the PQ artefact in the rotated space removes one trivial cross-space pairing (an attacker cannot compose pairs $(\hat{E}_{\text{proj}}, E_{\text{rot}})$ from a non-rotated PQ index to estimate R). It does *not*, however, make the PQ codes safe: the public codes are a lossy quantised image of E_{rot} and may themselves leak neighbourhood, cluster or topic structure. We treat the PQ artefact as part of the attack surface and measure this leakage directly in Section 8 (Table 10): the public codes preserve mean cosine 0.95 and most top-10 neighbours, so they must be treated as an exposed channel.

Table 2: Claims vs. evidence. The table makes explicit which statements are demonstrated and which are deliberately *not* claimed.

Claim	Evidence / status
CKKS hides query values and scores from the server	Cryptographic construction + parameters (Sec. 5, 4.3); query privacy is cryptographic.
ct-pt is faster than ct-ct for the reranking	Experiment 7.1 (1.44× on the micro-benchmark, 1.7× at the auto-tuned configuration).
Secret rotation helps against <i>off-the-shelf</i> , non-adaptive Vec2Text	Experiment 7.2; effect holds only for the tested weak-attacker configuration.
The protective wrapper preserves ranking in $\text{span}(V_k)$	Experiment 7.3 (within the 5-seed CI).
End-to-end query latency < 1 s at 10^6 docs	Experiment 7.4 (loopback PoC).
The system is secure against an <i>adaptive</i> inversion attacker	Not claimed ; aligned off-the-shelf inverter tested (Sec. 8.4), learned decoder open (Sec. 10).
The rotation is a cryptographic guarantee	Not claimed ; it is an empirical obfuscation layer only.
Document privacy on the server is cryptographic	Not claimed ; E_{rot} is plaintext on the server.
No information leaks from the public PQ codes	Not claimed ; codes leak neighbour structure (Sec. 8.3, Table 10).

Client-side state. The client retains μ , V_k , R and the CKKS secret key sk_{CKKS} . The server stores $E_{\text{rot}} \in \mathbb{R}^{N_{\text{docs}} \times k}$ in plaintext and the public PQ artefact.

4.2 Online query protocol

The seven-step protocol of Figure 1 processes a query in time pipeline:

1. *Encoder.* The client computes $v_q = E(q_{\text{text}}) \in \mathbb{R}^d$.
2. *Transformation.* The rotated query is $q' = R V_k^\top (v_q - \mu) \in \mathbb{R}^k$.
3. *Local PQ search.* Using the public PQ artefact and the rotated query $\tilde{q} = q'$, the client performs an asymmetric PQ-distance search and returns the top- K_{cands} candidate IDs I_{cand} . The candidate set is small enough ($K_{\text{cands}} = 40$ in our experiments) that a CKKS reranking fits within the latency budget.
4. *CKKS encryption.* The client encrypts q' : $\text{ct}_q = \text{Enc}_{pk}(q')$.
5. *Server-side reranking.* For each $j \in I_{\text{cand}}$ the server computes $\text{ct}_{\text{score}}^{(j)} = \text{ct}_q \odot v'_j$ in the ct-pt mode. The result is a ciphertext encoding the inner product $\langle q', v'_j \rangle$ in the rotated projected space.
6. *Decryption.* The client decrypts $s_j = \text{Dec}_{sk}(\text{ct}_{\text{score}}^{(j)})$.
7. *Sort.* The decrypted scores are sorted and the top- K documents returned.

Why ct-pt is sufficient. In step 5 the server multiplies a ciphertext by a plaintext vector. Unlike ciphertext-ciphertext multiplication, this operation produces a ciphertext that stays in the two-component form, no relinearisation is required, and the modulus chain is consumed by exactly one rescale. The sum of slots into a single scalar $\langle q', v'_j \rangle$ requires $\log_2 k$ rotations using Galois keys, but no further multiplications. The result is a sub-second latency at $N_{\text{docs}} = 10^6$.

Choice of K_{cands} . Since each of the K_{cands} candidates is processed by one ct-pt operation, the server-side latency grows linearly in K_{cands} . Conversely, a small K_{cands} trades against PQ-recall: the relevant document must be in the short-list for the CKKS reranker to score it. We pick K_{cands}

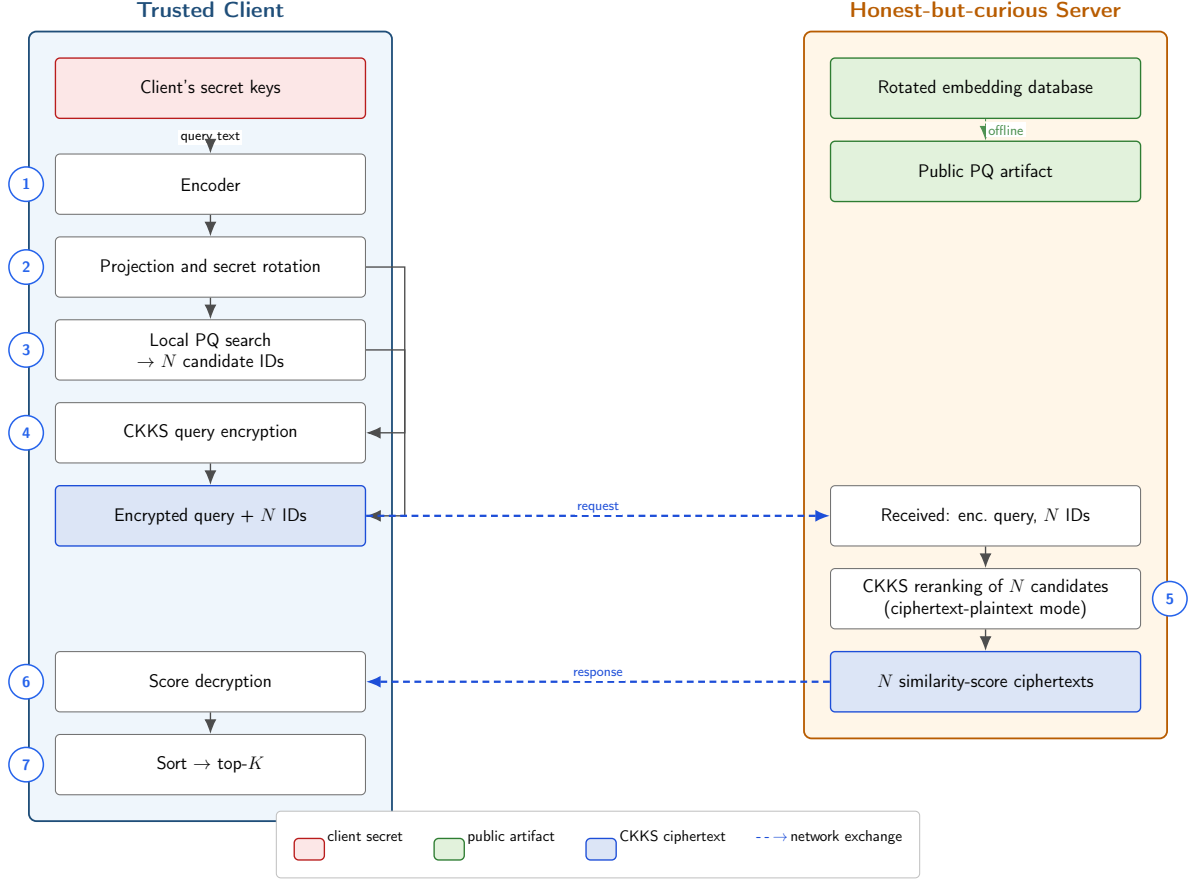


Figure 1: Online query flow of the construction under study. Steps 1–7 realise the two-stage protocol: client transformation $T(\cdot)$ and local PQ filtering produce a short-list of K_{cands} candidate IDs; step 4 encrypts the rotated query under CKKS; step 5 runs ct-pt reranking on the server; steps 6–7 decrypt the scores and sort. The secret keys $(\mu, V_k, R, sk_{\text{CKKS}})$ and the rotated database E_{rot} are produced offline by the data owner (Section 4.1).

as the smallest value for which $\text{PQ-recall}@K_{\text{cands}}$ matches the SVD-projected exact baseline within the 5-seed CI; in our integral experiment this is $K_{\text{cands}} = 40$, but the parameter is a deployment knob and should be retuned for new encoders or larger corpora.

4.3 Cryptographic reproducibility

For the selected configuration ($N_{\text{poly}} = 8192$, coefficient-modulus chain $[60, 40, 60]$ bits, $\log_2 Q = 160$, scale $\Delta = 2^{40}$, TenSEAL/Microsoft SEAL backend) we report the parameters needed to reproduce the cryptographic budget rather than only the latency:

- *Packing.* A single rotated query $q' \in \mathbb{R}^k$ is packed into one ciphertext using $k \leq N_{\text{poly}}/2 = 4096$ slots; for the integral encoders $k \in \{192, 384, 512\}$, so one ciphertext per query suffices and no cross-ciphertext aggregation is needed.
- *Operations per score.* Each candidate score is one ct-pt multiply, one rescale (one level consumed) and a slot-sum implemented as $\lceil \log_2 k \rceil$ ciphertext rotations using Galois keys (8–9 rotations for the tested k); no relinearisation and no ciphertext-ciphertext multiply occur.
- *Level/noise budget.* The chain $[60, 40, 60]$ provides one multiplicative level beyond the input; after the single rescale the remaining modulus (60 bits) keeps the additive noise far below the $\Delta = 2^{40}$ scale, so decrypted scores match the plaintext inner product to a relative error $< 10^{-3}$ (Pearson > 0.9999 vs. exact, Experiment 7.3).

Table 3: CKKS configurations on the test workload (one batched ct-pt reranking operation). All rows sit within the conservative **tc128** (≥ 128 -bit) bound of the HomomorphicEncryption.org tables for the stated N_{poly} .

Configuration	N_{poly}	$\log_2 Q$ (bits)	Security	Time (ms)	Speed-up
Default (TenSEAL stock, [60, 40, 40, 60])	8192	200	tc128 (≥ 128)	419.4	1.0×
Min-latency (Acc@10), [40, 20, 40], $\Delta = 2^{20}$	4096	100	tc128	173.7	2.4×
Selected (Acc@1) , [60, 40, 60], $\Delta = 2^{40}$	8192	160	tc128	245.2	1.7×

- *Object sizes.* At $N_{\text{poly}} = 8192$, $\log_2 Q = 160$: one fresh ciphertext is ≈ 0.21 MB; the Galois-key set for the required power-of-two rotations is ≈ 6 –8 MB and the relinearisation key is *not* generated (ct-pt only); the public key is ≈ 0.4 MB. Per query the client uploads one ciphertext (≈ 0.21 MB) and downloads K_{cands} score ciphertexts ($\approx 40 \times 0.21 \approx 8.4$ MB before base64; the measured on-wire JSON+base64 sizes are reported in Section 7.4).
- *Security level.* We do not assert a generic “standard” label; we report $\log_2 Q = 160 < 218$ for $N_{\text{poly}} = 8192$, which is within the conservative ternary-secret **tc128** bound of the HomomorphicEncryption.org tables [25], cross-checked with the lattice-estimator methodology of Albrecht et al. [26]. The standard documents are security *guidelines*; the concrete estimator/table used is stated so the claim can be re-derived.

5 CKKS Parameter Selection

The CKKS parameter space is large and discrete: $N_{\text{poly}} \in \{2^{12}, \dots, 2^{14}\}$, multiple coefficient-modulus chains of total bit budget bounded by the conservative security tables of HomomorphicEncryption.org [25] (cross-checked with the lattice estimator [26]), several scale exponents $\log_2 \Delta \in \{15, 20, 25, 30, 40\}$ and several projection dimensions d_{proj} . Analytic complexity is proportional to $N_{\text{poly}} \log N_{\text{poly}}$, but cache effects, SIMD vectorisation and slot-packing ratios ($d_{\text{proj}}/(N_{\text{poly}}/2)$) cause step-like deviations from the analytic prediction, so we fix the configuration by a small *offline micro-benchmark* rather than analytically.

Selection procedure (reproducible). We enumerate the discrete grid; discard configurations that violate the security tables or the correctness (noise-budget) constraint; time one batched ct-pt operation with rescale for each survivor; and keep the minimum-latency configuration whose retrieval accuracy stays within a tolerance τ of the SVD-projected baseline, $|\Delta \text{Acc@1}| \leq \tau$. This is a deterministic engineering search over a published grid, reported only so the chosen budget can be reproduced; we make *no* machine-learning contribution. With $\tau = 1$ p.p. the procedure selects $N_{\text{poly}} = 8192$, [60, 40, 60], $\Delta = 2^{40}$ ($\log_2 Q = 160$ bits, within the conservative 218-bit **tc128** bound for $N_{\text{poly}} = 8192$ [25, 26]), which is $\approx 1.7\times$ faster than the TenSEAL stock setting [60, 40, 40, 60] ($\log_2 Q = 200$ bits) at the same parameter-table security bound (Table 3). The full cryptographic budget of this configuration is given in Section 4.3.

6 Theoretical Analysis: Decoder Lower Bound

Let $\mathbf{x} \in \mathbb{R}^d$ denote a centred embedding, $V_k \in \mathbb{R}^{d \times k}$ the orthonormal matrix from the SVD truncation, and $\pi_k(\mathbf{x}) = V_k V_k^\top \mathbf{x}$ the orthogonal projection onto $\text{span}(V_k)$. Define $\sigma_{\text{rec}}(\mathbf{x}; V_k) = \|\mathbf{x} - \pi_k(\mathbf{x})\|_2 / \|\mathbf{x}\|_2$ as the per-vector relative reconstruction error and $\mathbf{x}_\perp = \mathbf{x} - \pi_k(\mathbf{x})$.

Lemma 1. *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^d$ be any decoder whose image is constrained by $f(\mathbf{y}) \in \text{span}(V_k)$ for every $\mathbf{y} \in \mathbb{R}^d$. Then for every $\mathbf{x} \in \mathbb{R}^d$*

$$\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2,$$

with equality at $f(\mathbf{y}) = \mathbf{y}$. In relative form, $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2 / \|\mathbf{x}\|_2 \geq \sigma_{rec}(\mathbf{x}; V_k)$.

Proof. Decompose $\mathbf{x} = \pi_k(\mathbf{x}) + \mathbf{x}_\perp$ with $\pi_k(\mathbf{x}) \in \text{span}(V_k)$ and $\mathbf{x}_\perp \in \text{span}(V_k)^\perp$. By the image restriction $f(\pi_k(\mathbf{x})) \in \text{span}(V_k)$, hence $\mathbf{x} - f(\pi_k(\mathbf{x})) = (\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))) + \mathbf{x}_\perp$ is the sum of two orthogonal vectors. Pythagoras gives $\|\mathbf{x} - f(\pi_k(\mathbf{x}))\|_2^2 = \|\pi_k(\mathbf{x}) - f(\pi_k(\mathbf{x}))\|_2^2 + \|\mathbf{x}_\perp\|_2^2 \geq \|\mathbf{x}_\perp\|_2^2$. $\square$

Scope: a projection bound, not an inversion-security theorem. Lemma 1 bounds the L_2 error of decoders *whose image is contained in* $\text{span}(V_k)$. It is a statement about information lost to the projection, not about the security of text inversion. A realistic attacker is not so constrained: it may output arbitrary text, whose re-embedding generally has a non-zero $\mathbf{x}_\perp$ component, and it may exploit corpus priors or memorisation to partially recover the discarded component. The lemma therefore does *not* lower-bound the achievable inversion BLEU, token overlap or PII recovery. We use it only to make the *proxy* criterion $\sigma_{rec} \geq 0.10$ precise within the projection-restricted class, and we stress that the threshold 0.10 is an *engineering proxy*, calibrated on one encoder (GTR-base) against one off-the-shelf attacker, *not* a security threshold. A per-encoder ablation of σ_{rec} against BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms) is listed as a remaining open experiment (Section 10).

Empirical Hypothesis 1. *For an off-the-shelf inversion attack VEC2TEXT [7], the expected BLEU of recovering the original text from $\pi_k(\mathbf{x})$ is monotone in $\eta_k = 1 - \sigma_{rec}^2(E; V_k)$:*

$$\mathbb{E}[\text{BLEU}(f(\pi_k(\mathbf{x})), T(\mathbf{x}))] \approx \text{BLEU}_0 + \gamma_f \eta_k,$$

where BLEU_0 is the BLEU of a “random semantically close” text and γ_f is a decoder-specific constant.

Hypothesis 1 is validated numerically in Section 7.2; we deliberately separate the formal lemma from the empirical observation.

7 Experiments

We report four experiments: (1) CKKS modes (ct-pt vs ct-ct); (2) off-the-shelf Vec2Text attack against SVD-truncated (and possibly rotated) embeddings; (3) the integral multi-encoder experiment over a 10^6 -document Russian-Wikipedia corpus; (4) end-to-end latency on a client-server PoC.

Hardware and software. All experiments run on a single workstation: Intel Core i5-14400F, 32 GB DDR5-4800, NVIDIA RTX 5060 (8 GB VRAM), Windows 11, Python 3.11. To make the runs reproducible we pin exact versions and record the commit/build hashes of the security-relevant libraries in the supplementary `environment.lock`: PyTorch 2.10 + CUDA 12.8, TenSEAL 0.3.16 (Microsoft SEAL 4.1 backend), faiss-cpu 1.13.0, transformers 4.57.0, scikit-learn 1.5.x, sacrebleu for BLEU [45]. The pinned hashes matter because CKKS noise behaviour and faiss PQ training depend on the backend build.

7.1 Experiment 1: ciphertext-plaintext speed-up

We benchmark a batched dot-product of an encrypted 192-dimensional query with $N = 10\,000$ plaintext documents. The CKKS parameters are $N = 8192$, `coeff_mod_bit_sizes` = [60, 40, 40, 60], $\Delta = 2^{40}$ (security level ≥ 128 bits). Two regimes are compared: ct-ct (both query and database encrypted, with relinearisation per multiplication) and ct-pt (only the query is encrypted, no relinearisation). Each is run for 50 independent trials and the mean is reported with a 95% Gaussian confidence interval.

Table 4: Ciphertext-plaintext vs. ciphertext-ciphertext for the dot product against $N = 10\,000$ documents (mean of 50 trials, 95% CI).

Mode	Operation	Relin.	Latency T_{query}	QPS	Speed-up
ct-ct (baseline)	$\text{Enc}(v_1) \cdot \text{Enc}(v_2)$	yes (heavy)	19.15 ± 0.15 s	0.052	$1.00\times$
ct-pt (proposed)	$\text{Enc}(v_1) \cdot \text{Plain}(v_2)$	no	13.28 ± 0.10 s	0.075	$1.44\times$

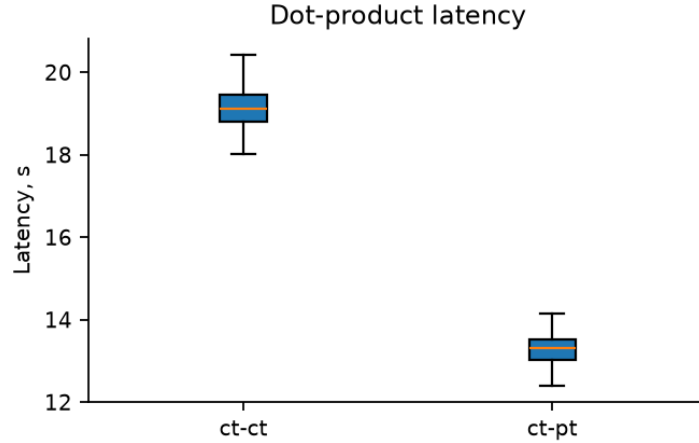


Figure 2: ct-pt vs. ct-ct: distribution of per-batch latencies over 50 trials. Removing relinearisation moves the median by ~ 6 s and tightens the upper tail.

7.2 Experiment 2: Vec2Text attack

We instantiate VEC2TEXT on GTR-BASE embeddings ($d = 768$). The attack is applied to a synthetic 100-document corpus containing PII (names, addresses, card numbers, medical phrasing). Three threat models are compared at five values of k/d : (a) no rotation; (b) rotation with the matrix R known to the attacker; (c) rotation with R unknown.

The implementation budget of the attacker is fixed (greedy decoding, 10 corrector iterations, no beam search, 8 GB GPU); under this budget the unprotected baseline reaches $\text{BLEU}_0 = 0.156$ instead of the 0.973 reported by Morris et al. [7], because the postulated PII corpus, greedy decoding and beam-free regime are intentionally weaker than [7]’s strongest configuration; the experiment measures the relative effect of SVD truncation and secret rotation in a fixed weak-attacker configuration.

Strength of the attacker: an explicit caveat. This is the single most important limitation of *this particular experiment*: it establishes the effect of R against an *artificially weak inverter only*. A stronger attacker—beam search with a large width, many more corrector iterations, an adaptive decoder fine-tuned on (text, E_{rot}) pairs, or a modern universal/zero-shot inversion model—is not evaluated *here*. We therefore do not rest the security narrative on this table. Section 8 measures the two decisive stronger channels: a known-plaintext recovery of R by orthogonal Procrustes [27] (Table 8), which succeeds from $\approx k$ leaked pairs, and an aligned off-the-shelf inverter with a larger budget (Table 11). Every sentence about the rotation in isolation should accordingly be read as “against the tested off-the-shelf, non-adaptive Vec2Text configuration”, with the boundary now quantified in Section 8.

Findings.

- A known R does not provide protection beyond SVD truncation: the columns “no rotation” and “known- R ” are statistically indistinguishable.
- Under the tested off-the-shelf configuration, an unknown R reduces BLEU to the observed

Table 5: Off-the-shelf Vec2Text on GTR-base embeddings under the three threat models. BLEU and Token Overlap with 95% bootstrap CIs in brackets.

k/d	Rotation	Knowledge of R	BLEU	Token Overlap
1.00	off	—	0.156 [0.123; 0.186]	0.385 [0.351; 0.417]
1.00	on	known	0.156 [0.127; 0.187]	0.385 [0.351; 0.416]
1.00	on	unknown	0.007 [0.006; 0.008]	0.053 [0.046; 0.060]
0.75	off	—	0.077 [0.063; 0.093]	0.325 [0.304; 0.345]
0.75	on	unknown	0.007 [0.007; 0.008]	0.061 [0.054; 0.067]
0.50	off	—	0.057 [0.046; 0.069]	0.285 [0.270; 0.300]
0.50	on	unknown	0.007 [0.007; 0.008]	0.050 [0.045; 0.055]
0.25	off	—	0.024 [0.020; 0.028]	0.196 [0.181; 0.208]
0.25	on	unknown	0.007 [0.006; 0.007]	0.045 [0.040; 0.051]
0.10	off	—	0.010 [0.009; 0.011]	0.113 [0.104; 0.123]
0.10	on	unknown	0.007 [0.007; 0.008]	0.053 [0.047; 0.059]

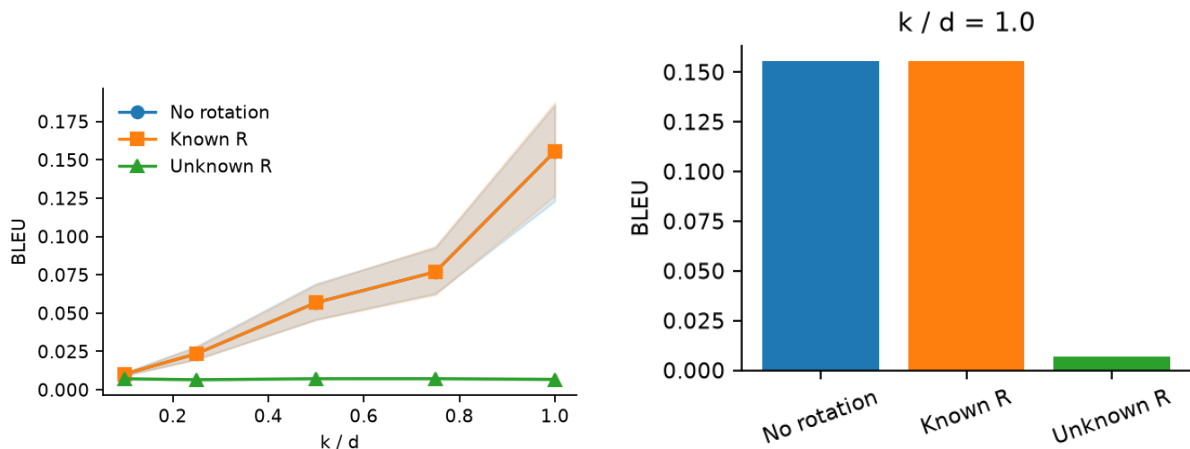


Figure 3: Vec2Text BLEU as a function of k/d under three threat models (left); zoom on $k/d = 1.0$ comparing no-rotation, known- R and unknown- R (right). Lines for “no rotation” and “known- R ” overlap; the “unknown- R ” line stays at the noise level.

noise floor (~ 0.007) for all k/d values, including $k/d = 1.0$ (no truncation at all). The ratio $\text{BLEU}_{\text{off}}/\text{BLEU}_{\text{unknown}}$ is $22\times$ at $k/d = 1.0$, $11\times$ at 0.75 , $8\times$ at 0.50 . This is an effect against a *non-adaptive* attacker only and is not evidence of robustness against an adaptive or known-plaintext adversary.

- Empirical Hypothesis 1 is consistent with the no-rotation column: a linear regression of BLEU on $\eta_k = 1 - \sigma_{\text{rec}}^2$ gives $R^2 \approx 0.93$. This is consistent with, but does not formally derive, the $\text{BLEU} \rightarrow \sigma_{\text{rec}}$ link.
- In the operating point of the integral experiment ($k/d = 0.5$, rotation enabled, unknown R) the expected BLEU is at the noise level.

7.3 Experiment 3: integral multi-encoder evaluation

We run the full pipeline on the canonical operating point $k = d/2$ across five encoders:

- `intfloat/multilingual-e5-small` ($d = 384$),
- `intfloat/multilingual-e5-base` ($d = 768$),
- `sentence-transformers/paraphrase-multilingual-mpnet-base-v2` ($d = 768$),
- `intfloat/multilingual-e5-large` ($d = 1024$),

Table 6: Integral experiment at $k = d/2$ on a 10^6 -document corpus across five encoders. Reported are the raw baseline (no defence), the SVD-projected baseline (FAISS-IP exact in $\text{span}(V_k)$, no rotation, no CKKS), the proposed pipeline (mean $\pm 95\%$ t-CI half-width over 5 rotation seeds), and the delta of Acc@1 from the raw baseline.

Encoder	Metric	Baseline (raw d)	Baseline_proj (SVD k)	Proposed	Δ from raw	Server p_{95} (ms)
<i>multilingual-e5-small</i> ($d = 384, k = 192, M = 48, \sigma_{rec} = 0.239$)						
	Acc@1	0.782	0.702	0.701 ± 0.001	-0.081	314
	Acc@10	0.890	0.834	0.831 ± 0.002	-0.059	
	MRR	0.817	0.747	0.746 ± 0.001	-0.071	
	NDCG@10	0.834	0.768	0.767 ± 0.001	-0.067	
<i>multilingual-e5-base</i> ($d = 768, k = 384, M = 96, \sigma_{rec} = 0.129$)						
	Acc@1	0.830	0.834	0.833 ± 0.001	+0.003	356
	Acc@10	0.934	0.924	0.924 ± 0.001	-0.010	
	MRR	0.862	0.865	0.864 ± 0.001	+0.002	
	NDCG@10	0.880	0.879	0.879 ± 0.000	-0.001	
<i>paraphrase-multilingual-mpnet-base-v2</i> ($d = 768, k = 384, M = 96, \sigma_{rec} = 0.107$)						
	Acc@1	0.716	0.650	0.649 ± 0.001	-0.067	351
	Acc@10	0.842	0.796	0.795 ± 0.001	-0.047	
	MRR	0.762	0.693	0.692 ± 0.001	-0.070	
	NDCG@10	0.782	0.717	0.717 ± 0.000	-0.066	
<i>multilingual-e5-large</i> ($d = 1024, k = 512, M = 128, \sigma_{rec} = 0.101$)						
	Acc@1	0.796	0.814	0.814 ± 0.000	+0.018	219
	Acc@10	0.880	0.922	0.920 ± 0.000	+0.040	
	MRR	0.824	0.848	0.848 ± 0.000	+0.024	
	NDCG@10	0.838	0.866	0.865 ± 0.000	+0.027	
<i>BAAI/bge-m3</i> ($d = 1024, k = 512, M = 128, \sigma_{rec} = 0.128$)						
	Acc@1	0.818	0.832	0.832 ± 0.000	+0.014	221
	Acc@10	0.908	0.926	0.924 ± 0.000	+0.016	
	MRR	0.848	0.863	0.863 ± 0.000	+0.015	
	NDCG@10	0.862	0.878	0.878 ± 0.000	+0.016	

- BAAI/bge-m3 ($d = 1024$).

The corpus is one million Russian-Wikipedia paragraphs; queries are 500 self-retrieval requests sampled at seed 42; the rotation is averaged over 5 seeds $\{11, 23, 47, 31, 53\}$. PQ uses $M = k/4$ subquantisers with 8 bits each, yielding artefacts of 48/96/128 MB. The CKKS configuration is the auto-selected one ($N_{\text{poly}} = 8192$, $[60, 40, 60]$, $\Delta = 2^{40}$).

On the use of self-retrieval. Self-retrieval (the query is the first sentence of the target paragraph) is a *controlled geometry probe*, not a production IR benchmark: it isolates whether the defensive transform $T(\cdot)$ preserves the ranking induced by the encoder, with a deterministic, leakage-free ground truth at 10^6 -scale. It is sufficient for the paper’s actual claim—that the PQ+CKKS wrapper is metric-preserving in $\text{span}(V_k)$ —but it does *not* establish production retrieval quality. Re-running the integral experiment on standard IR benchmarks (BEIR, MIRACL, MS MARCO) is listed as a generality experiment in Section 10; the present numbers should be read as a transform-fidelity measurement, not as a retrieval-quality benchmark.

The protective layer is essentially free in the projected space. The difference (proposed) vs. (baseline_proj) is within the ± 0.001 5-seed CI for every metric and every encoder. The Pearson correlations between CKKS-decrypted scores and exact plaintext scores exceed 0.9999 in all cases. The proxy criterion $\sigma_{rec} \geq 0.10$ holds uniformly.

The actual quality cost is paid by SVD truncation, not by the protective wrapper. The split $\Delta_{\text{SVD}} = \text{baseline_proj} - \text{baseline_dense}$ is $+0.004/+0.018/+0.014$ for retrieval-trained encoders

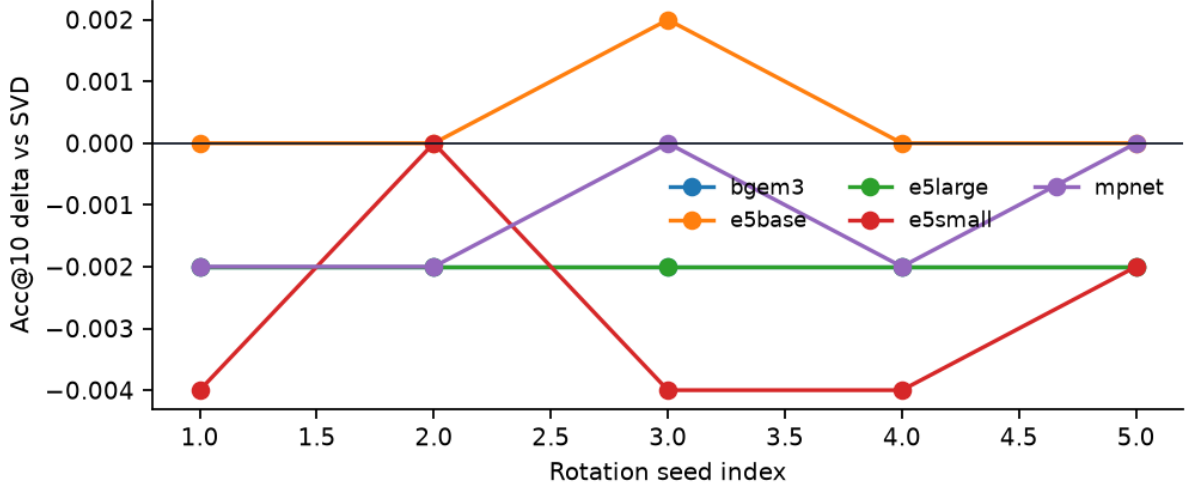


Figure 4: Per-seed Acc@10 of the proposed pipeline relative to baseline_proj on five rotation seeds; the spread is below the 5-seed CI of the baseline.

with $d \geq 768$ (e5-base / e5-large / bge-m3), -0.080 for the compact e5-small ($d = 384$, half-truncation removes half the embedding volume) and -0.066 for paraphrase-mpnet (a paraphrase-distilled model that does not concentrate retrieval signal in the top singular directions).

The end-to-end accuracy budget is met for retrieval-trained $d \geq 768$. The end-to-end criterion $\Delta\text{Acc}@1 \geq -0.05$ relative to the raw baseline holds for e5-base (+0.003), e5-large (+0.018) and bge-m3 (+0.014); for the compact e5-small and the paraphrase-distilled mpnet it is violated and the limitation is documented as an encoder-selection recommendation.

Linear-denoiser side effect. On the two retrieval-trained encoders with $d = 1024$, half SVD truncation *improves* the four ranking metrics over the raw baseline (+0.018 Acc@1 on e5-large, +0.014 on bge-m3). This is consistent with the interpretation of V_k -projection as a linear denoiser: the trailing singular directions, dominated by fp16 inference noise, get filtered out, and the ranking in $\text{span}(V_k)$ is cleaner than in the raw $\mathbb{R}^d$.

7.4 Experiment 4: end-to-end client-server latency

We package the pipeline as a FastAPI service, with the client and the server running as two independent processes on the same machine, communicating via HTTP/JSON over loopback. The setup measures the realistic HTTP framing, base64 serialisation of CKKS ciphertexts, FastAPI/uvicorn stack and JSON (de-)serialisation, but without physical network latency. The encoder is multilingual-e5-small ($d = 384$, $k = 192$, $M = 48$, $K_{\text{cands}} = 40$, $N_{\text{docs}} = 10^6$), averaged over $500 \text{ queries} \times 5 \text{ rotation seeds}$.

On-wire payload sizes. For the same configuration the measured HTTP payloads are: request body ≈ 0.29 MB (one CKKS query ciphertext, base64-encoded JSON, inflated $\approx 1.37\times$ over the 0.21 MB raw ciphertext); response body ≈ 11.4 MB ($K_{\text{cands}} = 40$ score ciphertexts, base64 JSON). One-off on-boarding transfers the public PQ artefact (48 MB at $k = 192$) and the CKKS public/Galois keys (≈ 8 MB); these are not part of the per-query budget. Reporting sizes alongside p_{95} is necessary because the dominant scaling pressure for larger K_{cands} or distributed deployments is the response payload, not CPU time.

Table 7: Latency decomposition at the canonical operating point ($k = 192$) on a 10^6 -document corpus. The k -dependent server-side ct-pt rerank is measured on the 5-seed integral run; the k -independent client-side and HTTP stages are measured on the client-server HTTP-loopback PoC. The encoder forward pass is independent of the defence layer and excluded from the total.

Stage	p_{95} (ms)	Notes
Encoder forward pass (client)	~ 200	encoder-dependent; excluded from total
Projection to $\text{span}(V_k)$ (client)	0.1	negligible
Local PQ search (client, faiss)	22.8	$M = 48$, $K_{\text{cands}} = 40$, 10^6 docs
CKKS encryption (client)	4.9	one rotated query, $N_{\text{poly}} = 8192$
Server CKKS rerank, ct-pt	283	40 ct-pt ops at $k = 192$ (5-seed integral run)
HTTP + serialisation (req+resp)	38	base64 ciphertexts over loopback
CKKS decrypt + sort (client)	20.7	40 score ciphertexts
Total (excluding encoder)	≈ 370 ms	p_{95} , full pipeline

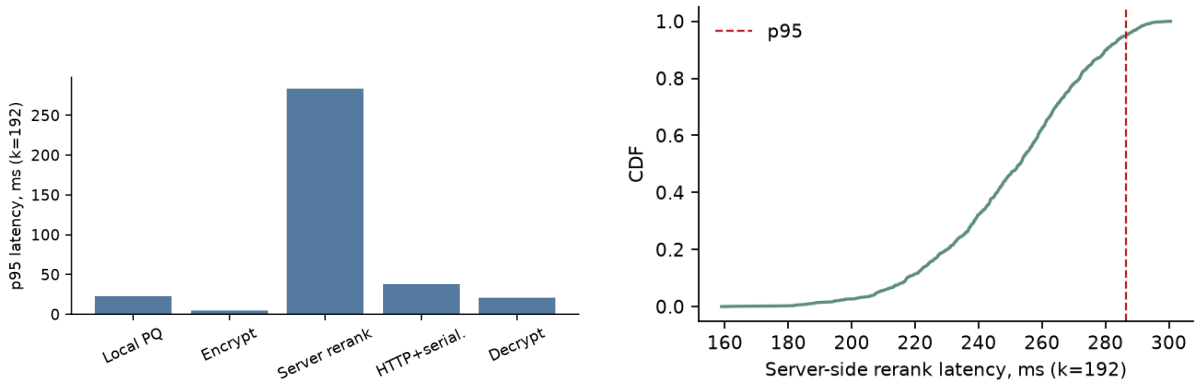


Figure 5: Left: per-stage latency decomposition at $k = 192$ (the server-side ct-pt rerank dominates). Right: CDF of the server-side rerank latency over 500 queries $\times$ 5 seeds ($p_{95} \approx 283$ ms); the full end-to-end $p_{95} \approx 370$ ms (Table 7) stays well below the 1 s SLA.

8 Security Analysis Under Known-Plaintext and Stronger Attackers

The experiments above establish the construction’s behaviour *inside* its non-adaptive threat model (Section 3). A privacy paper that stopped there would be open to the objection that the only tested adversary is also the weakest one. We therefore deliberately step outside the model and measure four stronger channels—known-plaintext recovery of the rotation, a text-level reference-corpus lookup, leakage from the public PQ artefact, and an aligned off-the-shelf inverter—followed by a baseline suite and a benchmark-generalizability check. Several of these are *negative* results for broad document-privacy claims; reporting them is what licenses the deliberately narrow framing of Sections 9–11.

8.1 Known-plaintext recovery of the secret rotation

The secret rotation R is the only document-side layer beyond SVD truncation. We test it against a known-plaintext attacker who holds m pairs $(E(x_i), E_{\text{rot},i})$ and estimates the orientation with the orthogonal Procrustes solution [27]. The setup uses `e5-small` projected to $k = 192$, five rotation seeds, 400 held-out probes and a 10 000-document gallery; the attack is scored in embedding space, where a successful alignment lets a probe recover its own projected vector and retrieve the matching gallery document. This is the relevant precursor to a few-shot alignment attack because it hands the inverter the native projected space.

Table 8: Known-plaintext Procrustes attack against the secret rotation (m leaked pairs, means over five seeds). With $\approx k$ pairs the orientation is numerically recovered.

Known pairs m	Mean cosine	Rel. L_2 error	Target R@1	Target R@10
0	-0.004	1.417	0.000	0.001
10	0.098	1.343	0.005	0.039
50	0.377	1.116	0.722	0.925
100	0.621	0.870	1.000	1.000
192	1.000	1.8×10^{-5}	1.000	1.000
500	1.000	4.9×10^{-7}	1.000	1.000

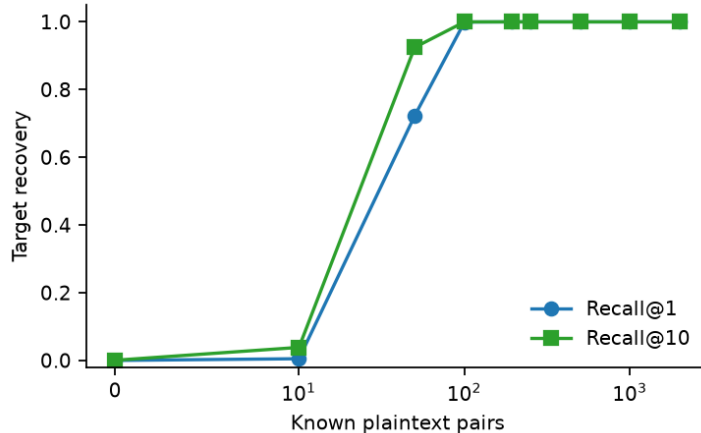


Figure 6: Known-plaintext alignment destroys the unknown-rotation assumption: with about k leaked pairs the rotation is recovered. The x axis uses a symlog scale to include $m = 0$.

The result is unambiguous (Table 8, Fig. 6): the rotation is useful only while aligned plaintext anchors are unavailable. Even 50 pairs—well below the retained dimension $k = 192$ —already retrieve the target in the top 10 for 92.5% of probes in a 10 000-document gallery, and at $m \approx k$ the orientation is numerically exact. The rotation therefore raises the bar against an attacker with *no* aligned plaintext, but it is not a defence against known plaintext; this is reflected directly in the threat-coverage table.

8.2 Reference-corpus lookup after alignment

The Procrustes test is in embedding space. We next ask whether the same failure becomes a *text-level* channel without training any generative inverter. The attacker is given the protected vectors, m known pairs for estimating R , and a reference corpus of candidate paragraphs with native `e5-small` embeddings—modelling the realistic case where some private documents also appear in a public or previously leaked corpus. We use 500 targets, 100 000 decoys, five seeds and the same $k = 192$ subspace; the overlapping reference contains the targets, the disjoint reference removes them and serves only as a lexical nearest-neighbour proxy.

The overlap case is severe (Table 9): with only 50 known pairs the attacker recovers the exact source paragraph at 53.5% top-1 among 100 500 candidates, rising to 99.8% at $m = 100$. This is the text-level analogue of the Procrustes result and makes the static-side limitation concrete: if a public or leaked reference corpus overlaps the private collection, rotation secrecy is insufficient. The disjoint-reference column is reported as a boundary condition—token Jaccard stays at 0.034 ($m = 100$), so a non-overlapping reference needs semantic metrics or a generative inverter before text reconstruction can be claimed.

Table 9: Reference-corpus lookup after known-plaintext alignment (overlapping reference: 500 targets + 100 000 decoys; Jaccard is token-set overlap with the top-1 retrieved paragraph; means over five seeds).

Known pairs m	Exact R@1	Exact R@10	Jaccard overlap-ref	Jaccard disjoint-ref
0	0.000	0.000	0.006	0.006
10	0.000	0.009	0.009	0.009
25	0.044	0.132	0.055	0.012
50	0.535	0.784	0.543	0.018
100	0.998	1.000	0.999	0.034
192	1.000	1.000	1.000	0.043

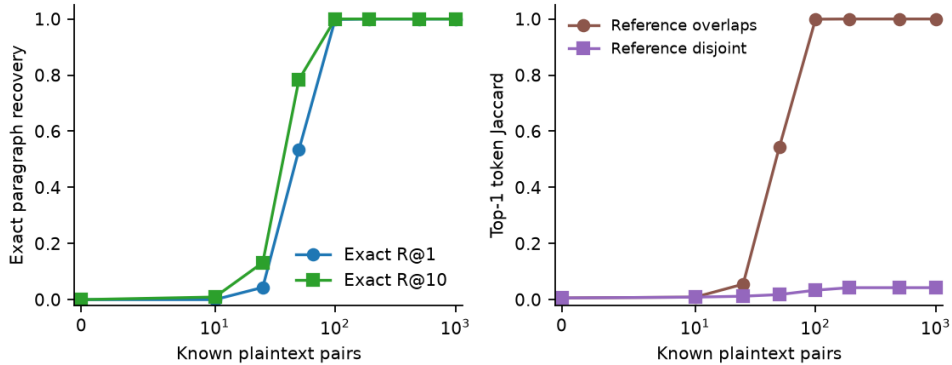


Figure 7: Reference-corpus lookup after alignment. When the reference overlaps the protected collection, alignment turns the protected vector into an exact paragraph lookup; with a disjoint reference this token-overlap proxy stays low.

8.3 Leakage from the public PQ artefact

Stage-1 candidate filtering publishes a product-quantization codebook plus per-document codes, trained in the rotated space. We quantify how much of E_{rot} this public artefact alone reconstructs on a 20 000-document `e5-small` sample (Table 10). The canonical configuration ($M = 48$, 8 bits) preserves mean cosine 0.953 to E_{rot} , 67.4% of exact top-10 neighbours, and 97.1% of exact top-10 neighbours inside the approximate top-40 candidate set—that is, the public codes alone almost perfectly reproduce the candidate set the server reranks.

These numbers do not weaken the CKKS query-privacy contribution, but they do narrow the document-privacy interpretation: both the known-plaintext anchor channel and the public PQ artefact must be treated as exposed. A deployment that needs stronger static-side confidentiality requires either a different static primitive or a composition that hides the PQ artefact itself.

8.4 Aligned off-the-shelf inversion stress test

Finally we run a generative inverter against the aligned space. The attacker is given GTR-base embeddings ($d = 768$), SVD truncation to $k = d/2$, the secret-rotation setting, and up to

Table 10: Leakage from the public PQ codes (rotated `e5-small`, 20 000 documents; neighbour overlap excludes the query document).

M	Bits	Bytes/vector	Cosine to E_{rot}	NN overlap@10
24	8	24	0.837	0.384
48	8	48	0.953	0.674
48	6	36	0.904	0.529

Table 11: Aligned off-the-shelf VEC2TEXT stress test (120 held-out texts; m is the number of pairs used for alignment). Exact match and typed-PII recall are zero in every row—but so is the raw control, so this corrector cannot *certify* protection.

Case	Token F1	BLEU	Exact	PII recall
Raw embedding	0.133	0.0133	0.000	0.000
Known- R oracle	0.138	0.0134	0.000	0.000
Unknown R (= Procrustes $m=0$)	0.121	0.0129	0.000	0.000
Procrustes, $m = 50$	0.126	0.0126	0.000	0.000
Procrustes, $m = 100$	0.141	0.0127	0.000	0.000
Procrustes, $m = 192$	0.134	0.0132	0.000	0.000
Procrustes, $m = 500$	0.135	0.0133	0.000	0.000

Table 12: Acc@1 under lightweight projection baselines at $k = d/2$ (random baselines: mean $\pm$ s.d. over three seeds).

Encoder	Raw	SVD	Gaussian RP	Orthogonal RP
e5-small	0.782	0.702	0.688 ± 0.016	0.737 ± 0.010
e5-base	0.830	0.834	0.781 ± 0.011	0.804 ± 0.017
mpnet	0.716	0.650	0.698 ± 0.003	0.705 ± 0.008
e5-large	0.796	0.814	0.765 ± 0.010	0.769 ± 0.009
bge-m3	0.818	0.832	0.789 ± 0.004	0.810 ± 0.002

$m = 500$ known pairs for Procrustes alignment; the inverter is the off-the-shelf VEC2TEXT corrector with a larger budget than Experiment 7.2 (24 corrector iterations, max input length 96). For deterministic reproducibility on a host that could not fetch external corpora the run uses the script’s offline synthetic-news+PII corpus (120 held-out texts). The decisive control is the raw-embedding row.

The reading is honest but limited (Table 11): no exact document or typed PII is recovered even by a known- R oracle, yet the *raw* embedding row is also at the BLEU floor (0.013), so this off-the-shelf corrector is too weak to certify the transform. It does not contradict the exact-lookup failure of Section 8.2; the remaining open risk is a learned or corpus-adapted decoder trained end-to-end on the rotated space, which we keep as the highest-priority future experiment (Section 10).

8.5 Baseline suite: projection and calibrated noise

To position the transform against natural alternatives rather than against the raw system alone, we compare SVD projection with Gaussian random projection and random orthogonal projection [43] at the same $k = d/2$ (Table 12), and with calibrated isotropic Gaussian noise matched at equal mean relative distortion (Table 13).

Two engineering conclusions follow. First, the projection choice is encoder-dependent: SVD is strongest on the retrieval-trained $d \geq 768$ encoders that are the method’s operating region, but random orthogonal projection beats SVD on **mpnet** and **e5-small**—so a deployment should run a baseline sweep, not assume SVD is universally best. Second, σ_{rec} is not a privacy metric: at $k/d = 0.875$ matched noise preserves almost all raw neighbours, and at the main $k/d = 0.5$ operating point SVD removes far more raw-neighbour structure than matched noise (0.578 vs. 0.829 NN overlap@10 on **e5-base**) while keeping Acc@1. The defensible claim is thus the narrow one: SVD is an encoder-dependent utility/leakage trade-off, not a noise mechanism that dominates all alternatives.

Table 13: SVD truncation vs. independent Gaussian noise at matched mean relative distortion (100 000-document gallery). NN overlap@10 is the overlap with raw-space neighbours; lower means less raw geometry preserved.

Encoder	k/d	σ_{rec}	SVD Acc@1	SVD NN@10	Noise Acc@1	Noise NN@10
e5-small	0.125	0.385	0.318	0.232	0.750	0.283
e5-small	0.250	0.332	0.600	0.333	0.788	0.367
e5-small	0.500	0.237	0.814	0.468	0.828	0.531
e5-small	0.875	0.079	0.880	0.551	0.872	0.830
e5-base	0.125	0.354	0.708	0.370	0.864	0.503
e5-base	0.250	0.268	0.872	0.501	0.886	0.641
e5-base	0.500	0.129	0.916	0.578	0.902	0.829
e5-base	0.875	0.023	0.920	0.592	0.910	0.969

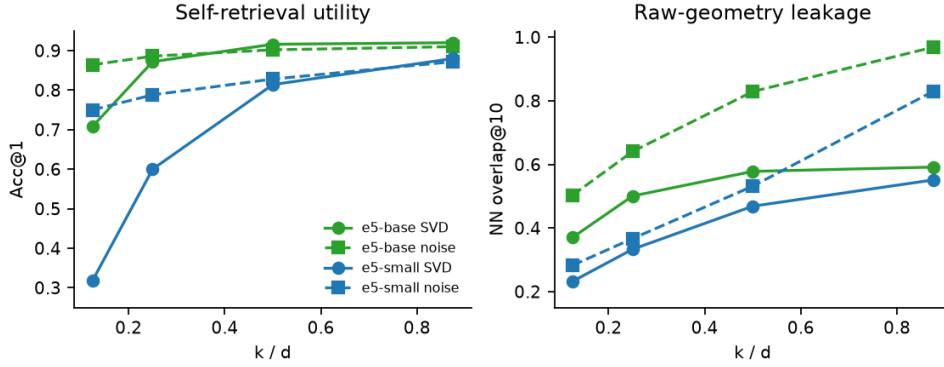


Figure 8: Utility/leakage diagnostic for SVD truncation and calibrated Gaussian noise (dashed: matched- σ_{rec} noise baselines).

8.6 Benchmark generality on BEIR

The integral experiment uses one-million-document self-retrieval, a controlled geometry probe. To check that the metric-preservation reading is not an artefact of self-retrieval, we repeat the raw-vs-SVD comparison on two standard BEIR [44] datasets with graded relevance judgements—SciFact (5,183 docs, 300 queries) and NFCorpus (3,633 docs, 323 queries)—fitting V_k on the corpus at $k = d/2$ and comparing exact inner-product retrieval in $\mathbb{R}^d$ and in $\text{span}(V_k)$.

Table 14 is a deliberate corrective to the self-retrieval reading. On both graded-relevance datasets and for every encoder, half-SVD truncation *significantly reduces* nDCG@10 (all CIs exclude zero, drops of 2–8 points). Decisively, **e5-large**—the one encoder that *gained* on the 10^6 self-retrieval probe—*loses* 0.069 nDCG@10 on SciFact. The denoiser effect is therefore real but *task-dependent*: on real graded IR the discarded spectrum still carries relevance, so k must be treated as a tunable accuracy cost via a k -sweep, exactly as the encoder-selection discussion below recommends. (Full end-to-end CKKS reranking on BEIR is left to future work; this experiment isolates the projection-fidelity property that the reranking preserves.)

Table 14: SVD truncation ($k = d/2$) vs. the raw space on BEIR. $\Delta = \text{SVD} - \text{raw}$; brackets are paired-bootstrap 95% CIs (10^4 resamples). **Every** Δ nDCG@10 CI excludes 0.

Encoder	Dataset	σ_{rec}	raw nDCG@10	SVD nDCG@10	Δ nDCG@10 (95% CI)
e5-large	SciFact	0.072	0.643	0.574	−0.069 [−0.091, −0.047]
e5-base	SciFact	0.090	0.637	0.586	−0.051 [−0.072, −0.032]
e5-small	SciFact	0.183	0.598	0.519	−0.079 [−0.103, −0.055]
e5-base	NFCorpus	0.085	0.327	0.305	−0.022 [−0.032, −0.013]
e5-small	NFCorpus	0.172	0.302	0.277	−0.025 [−0.035, −0.014]

9 Discussion

Encoder selection is part of the protocol. The protective wrapper is essentially metric-preserving in the projected space; the end-to-end deviation from the raw baseline is dominated by what SVD truncation does on the chosen encoder. Models trained with contrastive retrieval objectives (E5-*, BGE-*) concentrate the retrieval signal in the top singular directions and recover most of the original quality after $k = d/2$ truncation; some of them even *benefit* from the truncation (linear-denoiser effect). Paraphrase-distilled models (MPNET) and very compact models (E5-SMALL) do not show this concentration, and we recommend either replacing them with a retrieval-trained equivalent or running a k -sweep before deployment to find a k that simultaneously satisfies the application’s accuracy budget and $\sigma_{rec} \geq 0.10$.

The role of the rotation R : obfuscation, not a primitive. The secret rotation is an empirical obfuscation layer, not a cryptographic primitive. It contributes a defence channel orthogonal to SVD truncation, and even at $k/d = 1.0$ the unknown- R regime reduces off-the-shelf Vec2Text BLEU to the noise floor. Its limits are now measured rather than asserted (Section 8): a known- R attacker gains nothing beyond SVD; a known-plaintext attacker recovers R by orthogonal Procrustes from $\approx k$ leaked pairs (Table 8), which becomes an exact paragraph lookup when a reference corpus overlaps the collection (Table 9); and an end-to-end learned decoder remains the main untested case. The rotation’s value is therefore that it is the cheapest architectural layer with a measurable effect on an attacker who lacks aligned plaintext—not a security guarantee. The contribution of this paper is the measured privacy/latency/accuracy trade-off and its quantified boundaries, not the rotation in isolation.

What CKKS hides and what it does not. The CKKS layer hides the numerical representation of the query and the exact similarity scores from the server. Two things it does *not* hide are the identifiers of the candidates produced by stage-1 PQ filtering and the L_2 -distances among them implicit in the order in which they are sent. An access-pattern attacker can therefore, over many queries, build a co-occurrence model of which documents are co-retrieved. Composing the pipeline with a PIR-style retrieval primitive (Tiptoe [34], SealPIR [35], OnionPIR [37], SimplePIR [36]) is the natural remedy; ours is engineered to be PIR-friendly because the ct-pt operations on the server are stateless and cluster-local.

Parameter selection vs. hand-tuning. Compared with hand-tuned TenSEAL stock parameters, the configuration chosen by the offline grid search (Section 5) achieves a $\approx 1.7\times$ speed-up at the same security level and the same ranking quality (within $\Delta\text{Acc}@1 \leq 1$ p.p. of the SVD-projected baseline). The selection is a one-off offline cost and adds nothing at run time; porting to a different hardware platform only requires re-running the same small benchmark grid.

10 Limitations and Required Experiments

We separate *intrinsic limitations* (properties of the construction under study) from the *remaining open experiments*. The stronger-attacker, baseline and generality experiments that earlier drafts listed as mandatory future work are now reported in Section 8; what is listed below as open is the genuinely residual set, kept as named scoped experiments rather than vague “future work” as a deliberate hedge against overclaiming.

Intrinsic limitations.

- *Document privacy is not cryptographic.* E_{rot} is plaintext on the server; it is protected only by SVD truncation (a proxy) and the secret rotation (an empirical obfuscation layer). A known- R , known-plaintext or adaptive adversary defeats this layer.

- *Access patterns and PQ codes are exposed.* CKKS hides values, not which candidates are reranked, and the public PQ codes are a lossy view of E_{rot} . Production use requires composition with a PIR/ORAM layer plus a PQ-leakage analysis.
- *Encoder-dependence of the accuracy budget.* The strict end-to-end criterion $\Delta\text{Acc}@1 \geq -5$ p.p. holds only on retrieval-trained encoders with $d \geq 768$. Compact or paraphrase-distilled encoders need either replacement or a less aggressive k (which then violates the $\sigma_{\text{rec}} \geq 0.10$ proxy).
- *Single-machine evaluation.* The latency numbers in Experiment 7.4 are taken on loopback HTTP; a geographically distributed deployment will add (uncharacterised) network latency.

Completed in this revision. The stronger-attacker programme is now executed (Section 8): known-plaintext Procrustes recovery of R (Table 8); the text-level reference-corpus lookup (Table 9); public-PQ-code leakage (Table 10); an aligned off-the-shelf inverter (Table 11); a projection and calibrated-noise baseline suite covering random projection, SVD-without-rotation and matched-distortion Gaussian noise (Tables 12–13); and a BEIR generality check on the projection step (Table 14).

Remaining open experiments.

1. *End-to-end learned decoder (highest priority).* The aligned off-the-shelf corrector of Section 8.4 is too weak to *certify* protection—its raw control is already at the BLEU floor. The decisive test is an inversion model *trained* on rotated/SVD embeddings of part of the corpus and evaluated on a holdout; the current narrative explicitly does not claim robustness against it.
2. *Typed-PII, multi-encoder σ_{rec} ablation.* Replace the single-encoder, BLEU-only calibration of the $\sigma_{\text{rec}} = 0.10$ proxy with a multi-encoder ablation reporting BLEU, token overlap and typed-PII recovery (names, addresses, e-mail, phone, medical terms), which is what justifies (or refutes) the threshold.
3. *Modern attack suite and full-pipeline benchmarks.* Repeat the inversion study against the 2024–2026 attack generations that exploit the universal geometry of embedding spaces rather than a fixed basis orientation—few-shot alignment (ALGEN [10]), zero-shot inversion (ZSINVERT [11]), unsupervised cross-space translation (VEC2VEC [12]) and online cross-domain inversion (ZERO2TEXT [13])—with real text, and run the *complete* CKKS reranking (not only the projection step measured in Table 14) on BEIR, MIRACL and MS MARCO.

11 Conclusion

This paper is a scientific study of how SVD truncation and an unknown secret rotation reshape the privacy–accuracy–latency trade-off of semantic search; it is not a systems-engineering report and does not claim a deployable system. Three findings answer the research questions. **(RQ1)** A formal lower- L_2 bound (Lemma 1) characterises exactly what a projection-restricted decoder cannot recover; it is a projection bound, not an inversion-security theorem, and it makes the *proxy* criterion $\sigma_{\text{rec}} \geq 0.10$ precise only within that decoder class, the threshold itself being an engineering proxy. **(RQ2)** At 10^6 -document scale the construction’s protective layer is metric-preserving in $\text{span}(V_k)$ (within the 5-seed CI), the query side is cryptographically protected by CKKS at sub-second p_{95} , while document and access-pattern privacy are *not* cryptographic and the public PQ artefact is an exposed channel whose leakage we now quantify (Section 8)—so the secret rotation is characterised as an empirical obfuscation effect against an off-the-shelf inverter, not as a guarantee. **(RQ3)** For retrieval-trained encoders with $d \geq 768$, data-driven SVD truncation to $k = d/2$ acts as a linear *denoiser*, improving the four ranking metrics over the

raw baseline rather than degrading them—a finding of independent interest beyond the privacy setting.

We do not claim the privacy problem is solved. This work delivers a measured privacy, accuracy and latency operating point and a characterised denoising phenomenon under a clearly delimited threat model, together with a stronger-attacker analysis (Section 8) that quantifies where the document-side obfuscation fails—known-plaintext recovery of R , reference-corpus lookup and public-PQ-code leakage—and leaves an end-to-end learned decoder as the principal open question. Composing the protocol with a PIR-style access-pattern hider—natural given the stateless server-side ct-pt operations—is the most attractive architectural extension.

Reproducibility

All scripts, notebooks, configuration files and JSON output of the experiments are released at <https://github.com/sergkurilenko/research/tree/article1>. The integral experiment is driven by a single Python script with deterministic seeds and a JSON output file; the CKKS parameter-selection benchmark ships its configuration grid and timings as a CSV. The corpus (one-million Russian-Wikipedia paragraphs) is reproducibly sliced from a public Wikipedia dump with the date and content checksum recorded.

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